INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

TEST KIT, PETROLEUM:

AVIATION FUEL CONTAMINATION

(PORTABLE)

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an aviation fuel contamination testing kit which is portable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6630

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SPECIFICATIONS

FEDERAL

A-A-2344	-	Wrenches, Adjustable.
A-A-53608	-	Hydrometers, Graduated Scale (with Thermometer, API Gravity).
L-P-535	-	Plastic Sheet (Sheeting); Plastic Strip; Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Co-polymer, Rigid.
O-E-751	-	Ether, Petroleum; Technical Grade.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-C-4150	-	Cases, Transit and Storage, Waterproof and Water-Vaporproof.
MIL-P-26514	-	Polyurethane Foam, Rigid or Flexible, for Packaging.
MIL-T-27730	-	Tape, Antiseize, Polytetrafluoroethylene, with Dispenser.
MIL-C-46168	-	Coating, Aliphatic, Polyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40-inch x 48-inch Pallets, Skids, Runners, or Pallet Type Base.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1298	-	Density, Specific Gravity, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method.
D 2276	-	Particulate Contaminant in Aviation Turbine Fuels.
D 3240	-	Undissolved Water in Aviation Turbine Fuels.
D 3830	-	Filter Membrane Color Ratings of Aviation Turbine Fuels.
D 3951	-	Standard Practices for Commercial Packaging.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The aviation fuel contamination test kit (hereafter called "kit") is a one-man, portable kit consisting of components and testing equipment with the capability of determining temperature and API gravity, free-water content, and particulate contaminants in aviation fuels.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Glass. Unless otherwise specified (see 6.2), glass or other easily broken material shall not be used except for optical reason, or when specifically requested or approved by the Government.

3.3.2 Plastics. Plastic materials shall be resistant to discoloration or attack by aviation fuel samples and shall not become opaque under prolonged exposure to sunlight and aviation fuel samples.

3.3.3 Foam. Foam materials, where used, shall be charcoal (black) in color and of a polyethylene foam type; resistant to oil samples. When tested as specified in 4.6.2.7, the density shall be sufficient to allow firm cut-outs for the equipment to be fitted properly and to protect all equipment from damage.

3.4 Performance characteristics.

3.4.1 Temperature and API gravity. The test kit shall have the capability of determining temperature and API gravity by the hydrometer method in accordance with ASTM D 1298. The test apparatus and components shall be in accordance with figure 1 and as specified herein.

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3.4.1.1 Hydrometer. The hydrometer shall be in accordance with A-A-53608, type II, class and style as specified by the temperature range and the API gravity range as cited by figure 1.

3.4.1.2 Hydrometer holder. The holder shall consist of transparent shell marked (silk screened) so as to identify the hydrometer, a protective cap, and protective foam inserts to protect seven hydrometers.

3.4.1.2.1 Shell. The shell with a bottom shall be fabricated out of a clear polymethyl-methacrylate plastic. The overall height of shell shall be 271 millimeter (mm) ±5 mm with an inside diameter (I.D.) of 87 mm ±4 mm and a minimum wall thickness of 3 mm.

3.4.1.2.2 Foam.

3.4.1.2.2.1 Cushion foam. The bottom of the hydrometer holder shall have a 26 mm ±2 mm layer of cushion foam in accordance with MIL-P-26514, type I, class 2 to protect the lower end of the seven hydrometers.

3.4.1.2.2.2 Support foam. The support foam shall be in accordance with PPP-C-1752, type I, class 2. The support foam sections shall have six holes evenly spaced on a 25 ±2 mm radius and one hole in the center as specified herein.

3.4.1.2.2.2.1 Lower support foam. The lower support foam shall be on top of the cushion foam and shall be 113 ±5 mm in depth. The holes shall be through the complete depth of the support foam and of the size to firmly restrain the hydrometer from any movement.

3.4.1.2.2.2.2 Upper support foam. The upper support foam shall adequately fill the remaining space in the shell to protect the seven hydrometers, but allow sufficient space for an internal protective cap as depicted in figure 2. The holes in the upper support foam shall be of the size to firmly restrain the smaller end of the hydrometers. The entrance of the holes of the mating surface to the lower support foam shall be bell-shaped for correct fit where the hydrometers taper-down. A handle or method shall be made to allow easy removal of the upper support foam so as to allow access to the hydrometers.

3.4.1.2.3 Protective cap. A protective internal cap as depicted in figure 2, shall be of a soft plastic and fit firmly to prevent any movement of the foam.

3.4.1.2.4 Marking. On the shell, above the lower support foam line, opposite each hydrometer, shall be marked the nominal API gravity range. The markings shall be transfer, silk screen or by any other method suitable for the use intended which is as permanent as the normal life expectancy.

3.4.1.3 Hydrometer test cylinder. The hydrometer test cylinder shall be of a clear polymethylpentene material and shall be applied with or without molded-in or silk-screened gradations. The cylinder shall have an I.D. of 45 ±3 mm and have an overall height of 200 ±5 mm. The base shall be of a black polypropylene material and of sufficient size to resist tipping.

3.4.1.4 Calculaide oil gravity computer. The computer shall correct API gravity readings at any temperature between 0 - 200 °F to API gravity at 60 °F. It also gives corresponding specific gravity for any API gravity of 29 thru 81 minimum and conversely. The accuracy and specifics shall be as specified by ASTM D 1298. The size of the computer shall be such as to meet the requirements of 3.5.4.

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3.4.2 Particulate contaminant. The test kit shall have the capability of determining particulate contaminants in aviation turbine fuels by indicating changes in fuel and the acceptability ("go" - "no go") of the fuel through color ratings in accordance with ASTM D 3830. It shall also have the capability for taking samples for evaluation at a POL test laboratory in accordance with ASTM D 2276. The test apparatus and components shall be in accordance with figure 1 and as specified herein.

3.4.2.1 Particulate testing apparatus. To maintain interchangeability and ensure components compatibility, the following items shall be as solely specified in figure 1, unless otherwise noted, filter forceps, quick release valve and plug, valve hose assembly sampler, remote grounded sampling hose assembly fuel color standards, vacuum syringe with 2-way valves, adapters as specified, contamination monitors, and match weight monitors.

3.4.2.2 Accessories. The following components shall be as specified herein for the support to the particulate contamination test (see 4.6.2.3) as required. Citing of a component under this specific test does not limit its use to this test only.

3.4.2.2.1 Sample bottle. The sample bottle shall be transparent, narrow-mouth, polyvinyl chloride (PVC) bottle with a polypropylene 38 mm screw (leak-proof) closure. It shall have a minimum of 1,000 milliliter (ml) capacity with graduated (100 ml through 1,000 ml) subdivisions of at least 50 mm by silk screening. The overall height (with cap) shall be 205 mm ±3 mm and the overall diameter shall be 90 mm ±2 mm.

3.4.2.2.2 Solvent dispenser bottle. The solvent dispensing bottle shall contain 250 ml and be of polypropylene material with a polypropylene dropping closure and cap. Bottle should dispense a fine, steady stream when squeezed. The spout cap for the tip is captive and closure is linerless, non-contaminating and leak-proof.

3.4.2.2.3 Solvent. When specified (see 6.2), solvent shall be provided. Solvent shall be a prefiltered petroleum ether in accordance with O-E-751.

3.4.2.2.4 Tape. Tape shall be an antiseizing tape in accordance with MIL-T-27730, size II, 1/2-inch wide by 260 inches long with dispenser.

3.4.2.2.5 Wrench. Wrench shall be crescent adjustable type, 8-inches in length, and shall be in accordance with A-A-2344.

3.4.2.3 Spares. The kit is supplied with a replacement plastic tube with male luer adapter for support of the valve hose assembly sampler as specified in figure 1.

3.4.3 Undissolved water content. The test kit shall have a capability of determining the free-water content by using the Aqua-Glo method and ASTM D 3240.

3.4.3.1 Water content testing apparatus. To maintain interchangeability and to ensure component compatibility, the following items shall be as solely specified in figure 1: Aqua-Glo series III free-water detector, with line cord, external battery pack, connector wire for battery pack, calibration kit, monitor, tweezers and free-water pads. Individual pads package shall stipulate manufacturing date.

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3.4.3.2 Accessories. The Aqua-Glo test apparatus is supplemental in its test procedures with the 1,000 ml bottle cited under 3.4.2.2.1, and noted as a particulate contaminant test accessory.

3.4.3.3 Spares. The kit is supplied with a replacement ultraviolet bulb for support to the Aqua-Glo detector.

3.5 Carrying case.

3.5.1 Case. When tested as specified in 4.6.2.9, all items comprising the kit shall be packaged in a case conforming to MIL-C-4150, type II, style 2, class A, variety HC, except those changes as specified herein.

3.5.1.1 Case size. When tested as specified in 4.6.2.9, the case shall be as shown in figures 2 and 3 and shall not exceed the dimensions as shown. It may be smaller providing there is consideration for sufficient cushioning material to provide shock and vibration protection to the contents of the case during transport. The overall height of the case should not exceed 16.63 inches and, when open full, the top half of the case should be fully supported by the handle and the two halves should be approximately level.

3.5.1.2 Catch assemblies. Catches shall be operable by personnel wearing gloves.

3.5.1.3 Case finish. Unless otherwise specified (see 6.2), the portion of the case normally painted internally and externally shall be cleaned, treated or painted in accordance with MIL-T-704, type P or G as applicable. Unless otherwise specified (see 6.2), top coat shall be camouflage green 383 conforming to MIL-C-46168. Unless otherwise specified (see 6.2), non-metallic cases having the color pigmentation compounded in the material, shall be color matched to color 34094 of FED-STD-595 and shall not require painting except for hardware.

3.5.1.4 Case test. Testing shall be in accordance with MIL-C-4150 except for those changes as specified in 4.6.2.9.

3.5.2 Inserts. Inserts shall be in accordance with PPP-C-1752, type I and may be formed by the usage of one layer or more. The cut-outs shall be such as to restrain equipment from movement or damage, yet allow sufficient area for removal of the equipment. The cut-out locations shall be similar to that shown in figures 2 and 3.

3.5.3 Storage chest. A chest for containing the consumables shall be provided in the top section of the kit case. The chest shall be fabricated of aluminum or a glass fiber reinforced plastic which shall be divided into three sections. One section shall be of such size to contain 48 contamination monitors, one to contain 24 match weight monitors, and one to contain 2 boxes of free-water pads of 50 per box. The overall dimensions shall be approximately 7 inches by 12-1/2 inches by 4-1/2 inches deep. The chest shall be furnished with a latch for the lid which shall prevent accidental opening of the lid. A label shall be attached to the inside of the chest top which shall identify the components of each section by nomenclature, NSN, and quantity. A layer of removal cushioning material shall be installed between the chest top and the components to prevent damage to the label or the monitors.

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3.5.4 Placard. The case shall have a .13 inch to .19 inch thick opaque white placard in accordance with L-P-535, composition A, type II, for dividing the two halves. The placard shall be hinge-mounted and shall cover the area, except that area where the hydrometer test cylinder and sample bottle extends from the lower half into the upper half (see figure 2). Attached to one side of the placard shall be the Calculaide API oil gravity computer (see 3.4.1.4). Instructions for operation of the computer and the free-water detector shall be on one surface of the placard and instructions for taking samples and conducting the test for particulate contamination, shall be depicted on the second surface of the placard. The placard shall conform with 5.5 of MIL-STD-1472.

3.6 Weight. The weight of the complete test kit shall not exceed 42 pounds.

3.7 Environmental requirements. The test kit shall conform to the following environmental requirements.

3.7.1 Operating climatic conditions. When tested in accordance with 4.6.2.6, the kit shall perform as specified in 3.4 in any ambient temperature from +120 to -25 °F with the following exception. The exception being that the free-water detector shall be tested in the temperature range from +120 to +33 °F.

3.7.2 Storage climatic conditions. When tested in accordance with 4.6 2.6, the test kit shall not be damaged by storage in any ambient temperature from +145 to -28 °F (climatic design type basic in accordance with MIL-STD-810).

3.8 Transportability. The kit shall be capable of withstanding the vibrational stress and shock encountered when transported by all modes of transportation with the exception of the free-water detector.

3.9 Instructions.

3.9.1 Loading plan. Each kit shall be provided with a loading plan on the underside of the placard as shown in figure 2. The plan shall depict the proper location and method of identification of each item.

3.9.2 Particulate contaminate test. The operational instructions for this test, pictorially with verbiage, shall be on the underside of the placard as shown in figure 2.

3.9.3 Hydrometer test. The gravity calculator and instructions for this procedure shall be on the upside of the placard as shown in figure 3.

3.9.4 Free-water content test. The operating instructions for this test will be on the upside of the placard as shown in figure 3.

3.10 Marking. Each kit shall be identified and marked in accordance with MIL-STD-130.

3.11 Human factors. As applicable the kit shall conform to the requirements of MIL-STD-1472, with particular attention to 4.0, 5.5 and 5.13 of the standard.

3.12 Workmanship. The case and all component parts shall be fabricated free from any defects which could affect reliability, strength, safety, or serviceability. The cushioning material installed in the case shall be uniformly molded, formed, or cut. All markings, labels and printed matter shall be legible and protected from the environmental elements.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article kit shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article kit shall be subjected to the tests marked "X" in column 1 of table II. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 2 of table II. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination.

TABLE I. Examination schedule.

First Article	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
		MAJOR	
X	X	101. Material not as specified.	3.3
X	X	102. Component missing or not as specified.	Figure 1
X	X	103. Dimensions not as specified.	3.4.1.2 thru 3.4.1.3, 3.4.2.2.1
X	X	104. Components broken or damaged so as to affect serviceability.	3.4.1.1, 3.4.1.2, 3.4.1.3, 3.4.1.4, 3.4.2.1, 3.4.2.2, 3.4.3.1
X	X	105. Case dimensions not as specified.	3.5.1.1 & Figure 3
X	X	106. Case finish not as specified.	3.5.1.3
X	X	107. Cushioning inserts malformed or not cut to fit.	3.5.2
X	X	108. Storage chest exceeds dimensional requirements.	3.5.3
X	X	109. Placard not as specified, incomplete or not legible.	3.5 4
X		110. Weight limits exceeded.	3.6
X	X	111. Procedural instructions missing or inadequate.	3.9.2 thru 3.9.4
X	X	112. Workmanship not as specified.	3.12
		MINOR	
X	X	201. Loading diagram not as specified, missing, incomplete or not legible.	3.9.1
X	X	202. Identification plate/markings missing or not as specified.	3.10

4.5.2 Tests.

4.5.2.1 Test schedule

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TABLE II. Test schedule.

First Article	Quality Conformance	Defect	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X		Temperature & API gravity.	4.5.2.2	3.4.1
X		Particulate contaminate.	4.5.2.3	3.4.2
X		Undissolved water content.	4.5.2.4	3.4.3
X	X	Waterproof & water-vaporproof.	4.5.2.5	3.5.1
X		Environmental	4.5.2.6	3.7
X		Transportability.	4.5.2.7	3.8, 3.3.3
X		Human factor.	4.5.2.8	3.11
X		Carrying case.	4.5.2.9	3.5.1

4.5.2.2 Temperature and API gravity. The test shall be conducted in accordance with ASTM D 1298 with a known test sample. Failure of any component related to this test or the kit to meet applicable precision criteria shall constitute failure of this test.

4.5.2.3 Particulate contaminate test. The test shall be conducted in accordance with ASTM D 3830 with known acceptable and nonacceptable test samples. Failure of any component related to this test or the kit to meet applicable precision criteria shall constitute failure of this test.

4.5.2.4 Undissolved water content. The test shall be conducted in accordance with ASTM D 3240 with a known test sample. Failure of any component related to this test or the kit to meet applicable precision criteria shall constitute failure of this test.

4.5.2.5 Waterproof and water-vaporproof. Close and latch the test kit case, direct a water spray through nozzles to cover the parting line between the top and bottom of the case, hinge and the carrying handle. The spray nozzle shall give uniform water distribution and shall have a flow capacity of not less than 1 gallons per minute (gpm) per nozzle at 40 pounds per square inch (psi) water pressure. Fogging nozzles shall not be used. Angle the spray to each of the six case surfaces as specified herein so that water strikes the case at an angle of between 40 to 90 degrees. Spray-time on each case surface shall not be less than 3 minutes continuous at 30 to 40 psi water-line pressure and between 40 to 75 °F water temperature. Evidence of any water leakage to the interior of the case shall constitute failure of the test.

4.5.2.6 Environmental. Unless otherwise specified (see 6.2), each sample test kit with all accessories and components packed/stored in the carrying case shall be subject to testing as specified herein.

4.5.2.6.1 High temperature. Testing shall be in accordance with MIL-STD-810, method 501.2, procedure I and II except as modified herein. Upon completion of each test as specified, the accessories shall be removed and verified to be in operating condition. Verification shall be by performing all required operational functions for each accessory. The packing material shall be inspected for brittleness, cracking and deterioration. A failure of the accessory functional tests on the packing material shall constitute failure of these tests.

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4.5.2.6.1.1 Storage. The test kit shall be subjected to a high temperature environment of 145 ± 2 °F for a period of 4 hours after an initial 6 hours soak at 120 °F, raising and lowering of temperature to be accomplished within a time period of 1 hour for a total cycle of 12 hours, therefore a total minimum testing time of 84 hours. Upon completion of the last cycle, complete the steps of procedure I.

4.5.2.6.1.2 Operational. The test kit, with case open, shall be subjected to a high temperature of 120 ±2 °F for a period of 2 hours after an initiated 4 hours soak at 91 °F, raising and lowering temperature to be accomplished within a time of 3 hours for a total cycle of 12 hours. Conduct this procedure for a minimum of three cycles and follow thru with the steps of procedure II.

4.5.2.6.2 Low temperature. Testing shall be in accordance with MIL-STD-810, method 502.2 procedures I and II except as modified herein. A failure of the accessory functional tests or the packing material shall constitute failure of this test.

4.5.2.6.2.1 Storage. The test kit shall be subjected to a low temperature environment of -28 ±2 °F for a period of 24 hours and tested in accordance with MIL-STD-810, method 502.2 procedure I - storage. The packing material shall be inspected for brittleness, cracking and deterioration.

4.5.2.6.2.2 Operational. Upon completion of examination under procedure I the temperature shall be increased to -25 °F, for an additional 24 hours. At the end of this period, while still at -25 °F, the accessories shall be removed and verified to be in operating condition. Verification shall be performing all required operational functions for each accessory with the exception of the free-water detector.

4.5.2.7 Transportability. Unless otherwise specified (see 6.2), the kit shall be tested for: a) vibration in accordance with the basic transportation category for equipment carried as secured cargo, cited under method 514.3 of MIL-STD-810 in the typical mission/field transportation scenario of 530 miles; and b) for shock, in accordance with the procedures covering the transit drop under method 516.3 of MIL-STD-810, except that under the test facility, the barrier receiving the impact shall be an 8-inch thick bed of dry sand. Upon completion of this test, each sample kit shall be examined for damage to the case and contents. Any evidence of permanent deformation, cracks or broken items shall constitute failure of this test.

4.5.2.8 Human factors. The kit shall be evaluated throughout all testing to determine compliance with 3.11. Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.9 Case drop test. The drop test shall be in accordance with MIL-C-4150 except that all tests shall be performed on an 8-inch thick bed of dry sand with a 50 pound test load from a height of 48 inches. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For purposes of inspection, the unit of product shall be a completed pack of a single kit prepared for shipment.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. Presence of one or more defects shall be cause for rejection. All completed pallet loads shall be examined for the following defects.

113. Kit components not fitted securely in designated spaces within carrying case (see 5.1).
114. Preservation of the testing kit not as specified (see 5.1).
115. Carrying-case clasps not properly latched (see 5.1).
116. Shipping container not as specified for levels A and B (see 5.2).
117. Palletized loads not on type of pallet specified for levels A, B, or commercial (see 5.3).
118. Kits not secured to pallets as specified for all degrees (see 5.3).
119. Marking illegible, incomplete, incorrect, or missing (see 5.4).

5. PACKAGING

5.1 Preservation (see 6.4). Kit components shall be placed securely in their designated locations within the case. Desiccant shall be placed inside the case, in the quantity required for a method II pack, in accordance with MIL-P-116. The case lid shall be closed and latched. One tamper-proof seal shall be affixed to each clasp, in such a manner that the clasps cannot be released without destroying the seals.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The case containing the testing kit shall be packed in a close-fitting box conforming to PPP-B-636, V3c. Box closure shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The case containing the testing kit shall be packed as specified in 5.2.1, except the box shall be class domestic.

5.2.3 Commercial. The case containing the testing kit shall individually be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), the cases, packed as specified in 5.2, shall be palletized in accordance with MIL-STD-147, load type I, bonding C, K, L, storage aid 1.

5.4 Marking.

5.4.1 Military. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, shipping containers shall be marked "FRAGILE - DO NOT DROP" and with the word "UP" with an arrow toward the top of the container. The letters shall be 1-inch high, and the arrow shall be not less than 2 inches in length and 1/2 inch in width. Special markings shall be in accordance with MIL-STD-129 except as specified herein.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, shipping weight and cube shall be shown on each container and on each palletized load, as applicable.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The testing kit is intended for determining specific gravity, free-water content, and solid contaminants of aviation fuels in the field.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required; the number of first article test kits required, and specified tests to be performed-when different from table II (see 3.2).
d. When glass or other easily broken material shall be used for the kit (see 3.3.1).
e. When solvent shall not be provided (see 3.4.2.2.3).
f. When color and finish is other than as specified (see 3.5.1.3).
g. When test kit is not to be environmentally tested (see 4.5.2.6).
h. When test kit is not to be tested for transportability (see 4.5.2.7).
i. When a protective coating is not to be used in depot storage (see 5.1).
j. Degree of packing required (see 5.2).
l. When palletization is required (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Degree of preservation. Degree of preservation has not been specified as the preservation specified is the minimum acceptable for any degree. For purposes of preservation/packing level marking the preservation shall be marked A.

6.5 Subject term (key word) listing.

Carrying case
Contaminants
Hydrometer
Particulate contaminants
POL
Solvent

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6640-A546

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FIGURE 1. Aviation Fuel Contaminant Test Kit. X-4109D

FIND NO.	QTY	DESCRIPTION	NSN	SPECIFICATION	POSSIBLE SOURCE OR EQUAL	
					VENDOR CODE (FSCM)	PART NO.
1	1	Case, Carry, Testing Kit with: Computer, Oil Gravity, API (Aviation Fuel) Instruction Placard Insert, Secure, Foam Material	6630-01-232-9531		54333 08071	100575-1 XX64037 02
2	1	Bulb, Replacement, Ultraviolet			32218	GTP-2380
3	1	Syringe, Vacuum With 2-Way Valves	6640-00-086-6326		08071	XX62000 35
4	1	Hydrometer 29-41 Gravity Range	6630-00-265-7758	A-A-53608, Type II, Class 2, Style 4	(58536) 32218 75296	GTP-1706 7606
5	2	Hydrometer 39-51 Gravity Range	6630-00-265-7759	A-A-53608, Type II, Class 2, Style 5	(58536) 32218 75296	GTP-1707 7608
6	2	Hydrometer 49-51 Gravity Range	6630-00-265-7764	A-A-53608, Type II, Class 2, Style 6	(58536) 32218 75296	GTP-1708 7610
7	1	Hydrometer 59-71 Gravity Range	6630-00-265-7765	A-A-53608, Type II, Class 2, Style 7	(58536) 32218 75296	GTP-1709 7612
8	1	Hydrometer 69-81 Gravity Range	6630-00-815-2267	A-A-53608, Type II, Class 2, Style 8	(58536) 32218 75296	GTP-1710 7614
9	1	Holder, Hydrometer, Gatling, for Seven Hydrometers	6635-01-232-9540		54333	100565-1
10	1	Bottle, Narrow Mouth, PVC, Graduated, 1,000ml Capacity, Screw Cap	6630-01-232-9537		08071 54333	XX64037 06 100573-1
11	1	Monitor, Stainless Steel	6630-01-232-9542		08071 32218	XH00000 06 GTP-3326
12	1	Bottle, Low Density Polyethylene, Solvent Dispenser, 250ml Capacity, Screw Cap with Tip and Tubing			05178 54333	2401-0250 100588-1
13	1	Quick Release Valve, Female, 1/8 NPTF and Plug	6630-00-009-1423		08071 54333	XX64037 35 100580-1
14	1	Tweezers (For Pads)	6635-01-232-9536		32218	GTP-293

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FIGURE 1. Aviation Fuel Contaminant Test Kit. (continued)

FIND NO.	QTY	DESCRIPTION	NSN	SPECIFICATION	POSSIBLE SOURCE OR EQUAL	
					VENDOR CODE (FSCM)	PART NO.
15	1	Forceps, Filter	6640-00-426-0300		08071	XX62000 06
16	1	Wrench, Crescent, Adjustable, 8-In	5120-00-240-5328	A-A-2344	(58536) (94480)	10-300
* 17	1	Detector Unit, Free Water, Aqua-Glo, Series III, with Instrument Pack GTP-768, Cable GTP-2402 with Ultra-violet Bulb and Calibrating Standard (PAD) GTP-0764	6640-01-235-7877		32218	GTP-2855
18	1	Sampler, Valve Hose Assembly	6630-00-999-0753		08071	XX64037 08
* 19	1	Jeweler's Screwdriver			32218 54333	GTP-765 100591-1
20	1	Remote Sampling Assembly	6630-00-999-0754		08071	XX64037 05
21	1	Battery Pack, External for Aqua-Glo	6140-01-232-9546		32218	GTP-2404
22	1	Connector Wire for External Battery	6630-01-232-9547		32218	GTP-2403
23	2	Free-Water Pads, (25mm) 50/Box	6640-00-235-3820		32218	GTP-25
24	24	Match Weight Monitors	6630-00-764-5761		08071	MAWP037PM
25	48	Contamination Monitors	6630-00-445-3662		08071	MAWP037PO
26	1	Standards, Fuel Color	6640-00-326-7684		08071	XX64037 85
27	1	Tape, Anti-Seize, 1/2-Inch X 260 Inches	8030-00-889-3535	MIL-T-27730, Size II	(81349)	
28	1	Cylinder, Hydrometer Test, 150ml			08071 54333	XX64037 09 100574-1
29	1	Nipple, Hex, 1/4 NPTM - 2 Inch Long			08071 54333	XX6700125 100581-1
** 30	1	Plug, Sampler, Thread W/O Ring			08071	XX64037 20
31	1	Valve, Quick Release, Male, 1/8 NPTF and Cap			08071 54333	XX64037 16 100583-1
32	1	Adapter, Reducing, 1/4 NPTF X 1/8 NPTM			08071 54333	XX64037 15 100585-1
33	1	Chest, Storage, Compartmentalize for Monitors/Pads			08071 54333	XX64037 100586-1

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FIGURE 1. Aviation Fuel Contaminant Test Kit. (continued)

FIND NO.	QTY	DESCRIPTION	NSN	SPECIFICATION	POSSIBLE SOURCE OR EQUAL	
					VENDOR CODE (FSCM)	PART NO.
34	1	Plastic Container w/Cap			08071 54333	XX64037 19 100586-1
*** 35	1	Plastic Tube w/Male Luer Adapter (Spare Part)			08071 54333	XX64037 21 100595-1
**** 36	1	Screw Closure, Red Nylon			05178	2165-0240
37	3	O-Ring (Low Temperature) (EA) (BAG of 3)			02697 08071	2-110-677-70 XX64037-17

NOTES:

* 1. Calibration pad supplied with detector unit, find no. 17, stored in container, find no. 34.
** 2. Find no. 30 is part of find no. 18 but vendor may furnish it as separate part or with find no. 20.
*** 3. Find no. 35 is a spare of item furnished with find no. 18.
4. For storage, items may be separated.
5. Variations in position of components is allowable.
**** 6. Find no. 36 is to be used as a closure to find no. 12, material shall be as specified or fluoride gas treated polypropylene. Storage slot is required for cap with tube that is supplied as part of find no. 12.
7. Find no. 37 stored in container, find no. 34.

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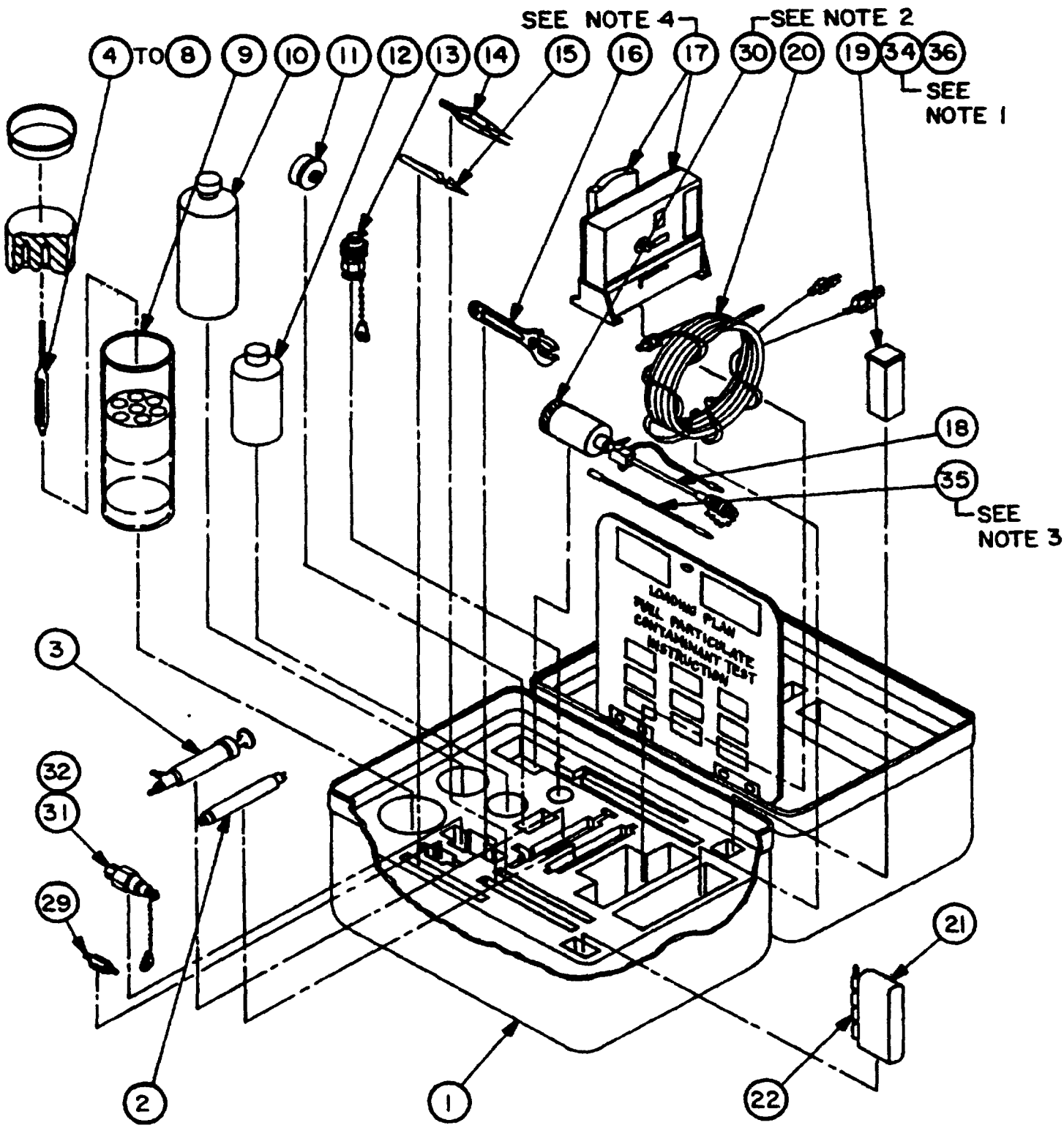


FIGURE 2. AVIATION FUEL CONTAMINANT TEST KIT

X-4110D

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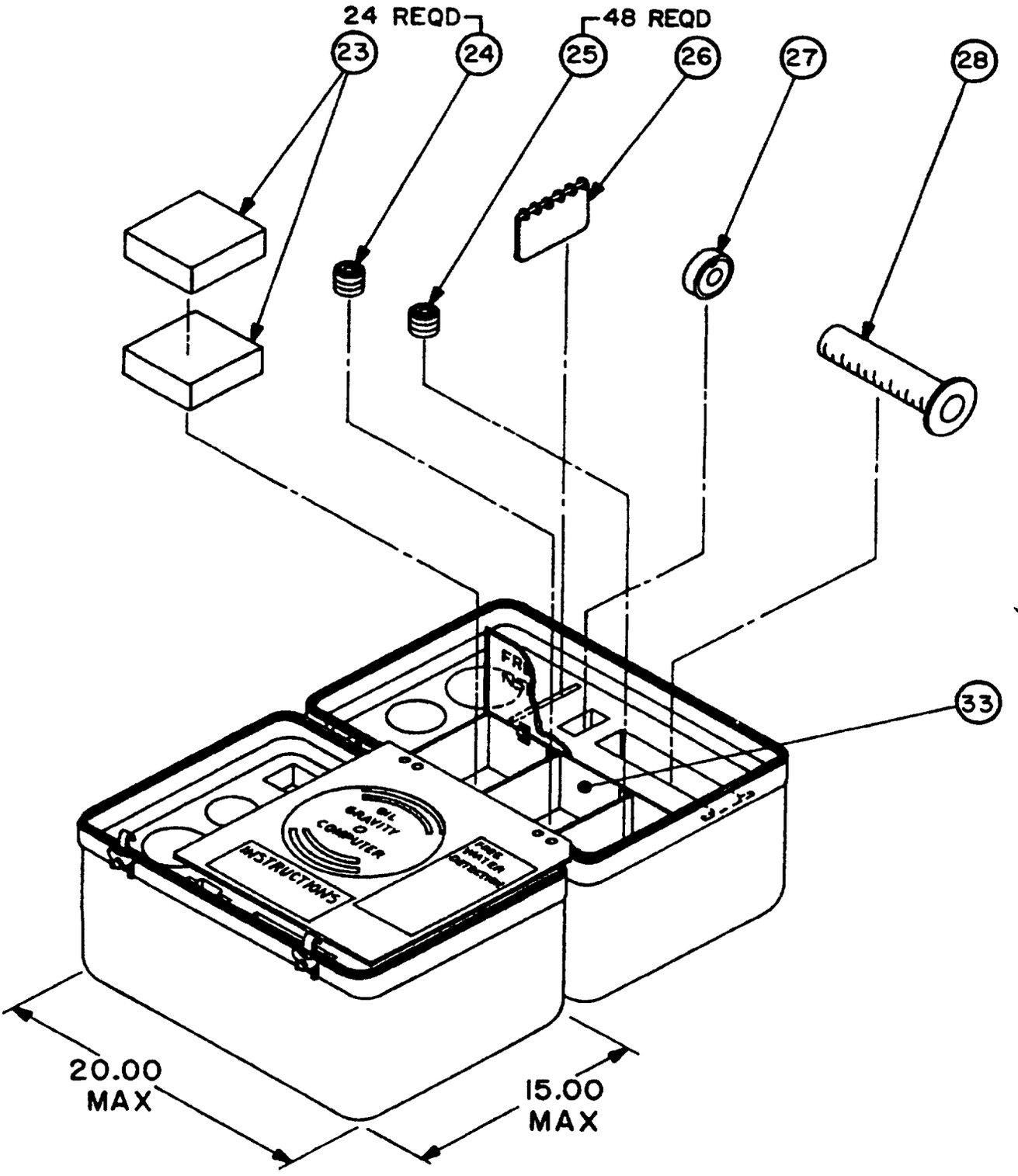


FIGURE 3. AVIATON FUEL CONTAMINANT TEST KIT

X-4111A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52849C(ME)	2. DOCUMENT DATE (YYMMDD) 931027

3. DOCUMENT TITLE
Test Kit, Petroleum: Aviation Fuel Contamination (Portable)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340	

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